

PIONEER JUNIOR COLLEGE, SINGAPORE

**JC2 PRELIMINARY EXAMINATION
HIGHER 2**

CANDIDATE
NAME

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CT
GROUP

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INDEX
NUMBER

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CHEMISTRY

9746/02

Paper 2 Structured

24 September 2009

1 hour 30 min

Additional Materials: Data Booklet

READ THESE INSTRUCTIONS FIRST

Write your name, CT Group and index number in the spaces at the top of this page.
Write in dark blue or black pen.

Answer **all** questions.
Write your answers in the spaces provided on the question paper.

You are advised to show all working in calculations.
You may use a calculator.
The number of marks is given in brackets [] at the end of each question or part question

FOR EXAMINER'S USE	
1	/ 14
2	/ 16
3	/ 10
4	/ 14
5	/ 6
TOTAL	
/60	

- 1 (a) Naturally occurring gallium, Ga, (relative atomic mass: 69.7) is a mixture of two isotopes, gallium-69 and gallium-71.

(i) What do you understand by the term *relative atomic mass*?

(ii) Using the information given above, calculate the percentage abundance of each isotope.

[3]

- (b) The boiling points of the chlorides of aluminium and phosphorus are given below:

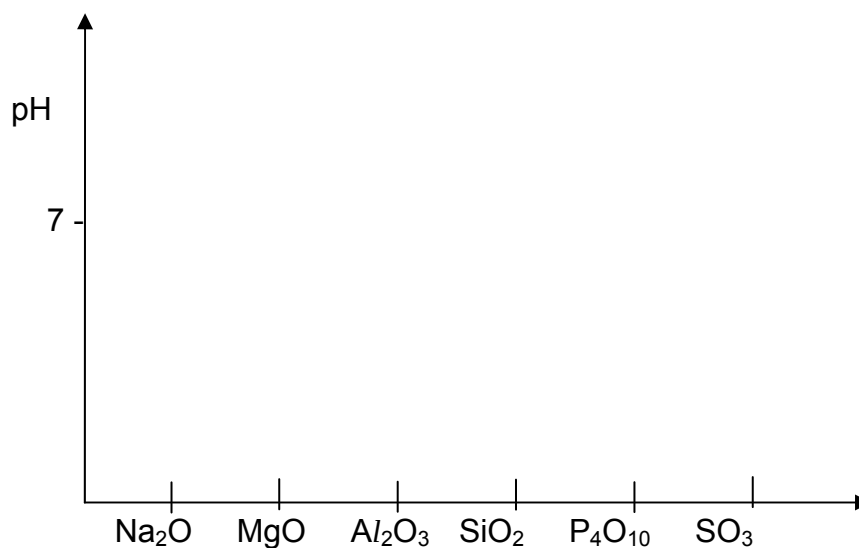
Compound	b.p./ °C
NaCl	1465
AlCl ₃	180
PCl ₅	160

Relate the boiling points of these three chlorides to their structure and bonding.

[3]

- (c) The highest oxides of the elements sodium to sulfur are added to separate beakers of water.

- (i) Using the axes provided below, sketch a graph to illustrate the variation of pH of the resulting mixtures.



- (ii) With the aid of equations (where appropriate), account for the two different pH values of Al_2O_3 and P_4O_{10} in your sketch.

[3]

- (d) (i) Dolomite is a double carbonate of calcium and magnesium, and has the formula $\text{CaMg}(\text{CO}_3)_2$. Both carbonates will decompose on heating.

State and explain which of the carbonates will decompose first on heating.

- (ii) Group II hydroxides and nitrates decompose to give the same solid residues but different gaseous products.

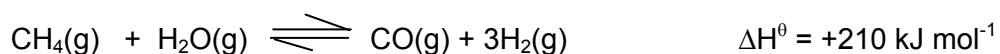
A Group II hydroxide exists as a hydrate, $\mathbf{M}(\text{OH})_2 \cdot n\text{H}_2\text{O}$. When a 1.575 g sample of this hydrate was heated, the mass of the sample decreased by 0.720 g. Subsequent heating to constant mass produced a further mass loss of 0.090 g.

Calculate the value of n , and deduce the identity of the metal $\mathbf{M}$.

[5]

[Total: 14]

- 2 (a) Hydrogen is manufactured by steam reforming of natural gas which involves the following reaction:



An equimolar mixture of $\text{CH}_4(\text{g})$ and $\text{H}_2\text{O}(\text{g})$, at total initial pressure of 2 atm, is allowed to react in the presence of nickel catalyst in a closed vessel at 1000 K. When equilibrium is attained, the partial pressure of CH_4 is found to be 0.57 atm.

- (i) Calculate the partial pressures of H_2O , CO and H_2 at equilibrium, and hence the new total pressure.

- (i) Write the expression for the equilibrium constant, K_p , and calculate its value.

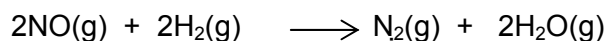
- (iii) State, with explanation, the effect of each of the following changes on K_p for the above reaction.

1. increasing the temperature from 1000K to 1500 K.

2. adding an inert gas, such as helium, into the vessel

[8]

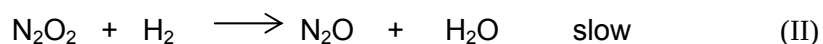
- (b) Hydrogen reacts with nitrogen monoxide to give nitrogen and steam as shown in the following equation. The following data were collected for the reaction between hydrogen and nitrogen(II) oxide.



Experiment	[H ₂]/ mol dm ⁻³	[NO]/ mol dm ⁻³	Initial Rate/ mol dm ⁻³ s ⁻¹
A	0.0100	0.0250	2.40 x 10 ⁻⁶
B	0.0050	0.0250	1.20 x 10 ⁻⁶
C	0.0750	0.0750	1.62 x 10 ⁻⁴

- (i) What is the rate equation for the reaction?

- (ii) The following mechanism was proposed for the reaction between NO(g) and H₂(g).

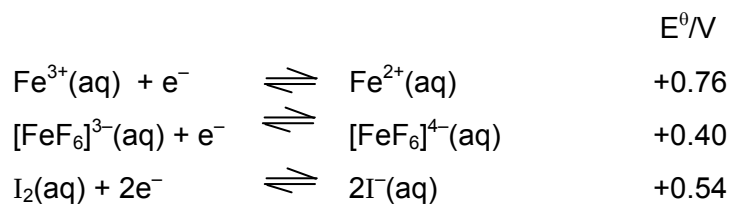


- Draw dot and cross diagrams to show the bonding of the two molecules NO and N₂O₂. Hence suggest why (I) is a likely first step in the reaction.
- Is the rate equation obtained in (b)(i) consistent with the proposed mechanism? Explain your answer.

[8]
[Total: 16]

3 Iron forms coloured compounds mainly in the +2 and +3 oxidation states.

(a) Consider the standard reduction potentials listed below:



Explain the following observations, giving a balanced equation in each case.

(i) When $\text{Fe}(\text{NO}_3)_3(\text{aq})$ is added to $\text{KI}(\text{aq})$, a brown solution is formed.

(ii) When concentrated $\text{KF}(\text{aq})$ is added to the resulting solution in **(a)** above, the brown colour fades.

[5]

- (b) (i) Iron(III) iodate(V), $\text{Fe}(\text{IO}_3)_3$, is a sparingly soluble ionic compound.
At 298 K, the solubility of $\text{Fe}(\text{IO}_3)_3$ is 3.6 g dm^{-3} . Calculate the K_{sp} of $\text{Fe}(\text{IO}_3)_3$ at 298 K.

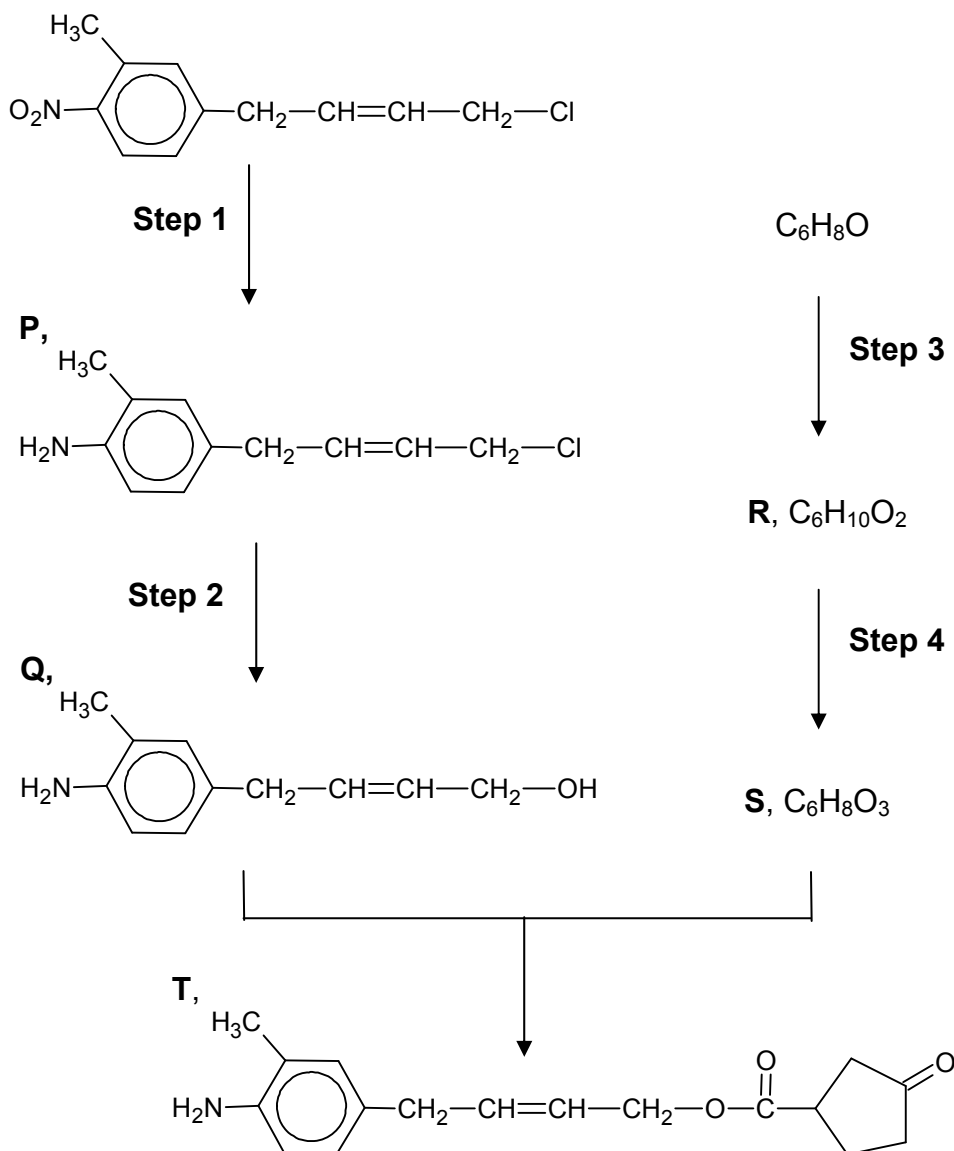
- (ii) To 1 dm^3 of $0.105 \text{ mol dm}^{-3} \text{ K IO}_3(\text{aq})$, solid $\text{Fe}(\text{IO}_3)_3$ is added until no more solid dissolves.

What is the solubility of $\text{Fe}(\text{IO}_3)_3$ in $0.105 \text{ mol dm}^{-3} \text{ K IO}_3$?

[5]

[Total: 10]

- 4 The diagram below shows the synthetic pathway by which compound **T** may be prepared.



- (a) Give the structural formula for compounds **R** and **S**.

R	S

[2]

- (b) Suggest reagents and conditions for steps **1** to **4**:

Step 1

Step 2:

Step 3:

Step 4:

[4]

- (c) Outline the mechanism for the conversion of **P** to **Q**.
(Note: You may use “R” to represent the part of **T** that does not take part in the reaction.)

[3]

(d) Draw the structures of the organic products formed when Compound **T** is reacted with each of the following.

(i) CH_3COOH

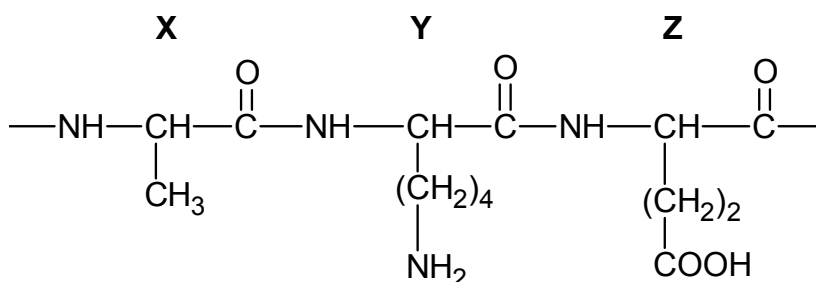
(ii) acidified potassium manganate (VII), heat

(iii) aqueous bromine

[5]

[Total: 14]

- 5 The diagram below shows the structure, at pH 7, of a fragment from a protein which contains three amino acids, **X**, **Y** and **Z**.



- (a) Draw the structures of the three amino acids, **X**, **Y** and **Z**, that could be obtained by hydrolysis of the protein fragment.

X	Y	Z

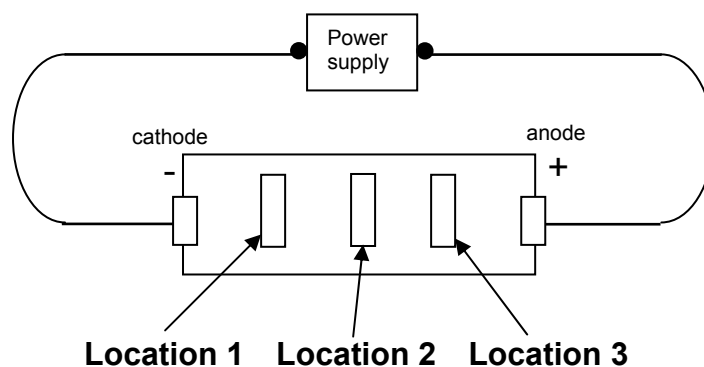
[3]

- (b) Given that the isoelectric point, IEP of the amino acids, **X**, **Y** and **Z** are 6.0, 9.7 and 3.2 respectively.

In order to separate the mixture containing the three amino acids, a student placed a streak of the mixture in the centre of a piece of filter paper wet with a buffer solution of pH 6.0.

The student then connected two electrodes to the edges of the paper and a potential difference was applied.

After some time, the student sprayed the filter paper with ninhydrin and the locations of the amino acids are shown in the diagram below.



By drawing the corresponding forms of the three amino acids in the buffer solution, draw the structures of the chemical species present.

Location 1
Location 2
Location 3

[3]

[Total: 6]